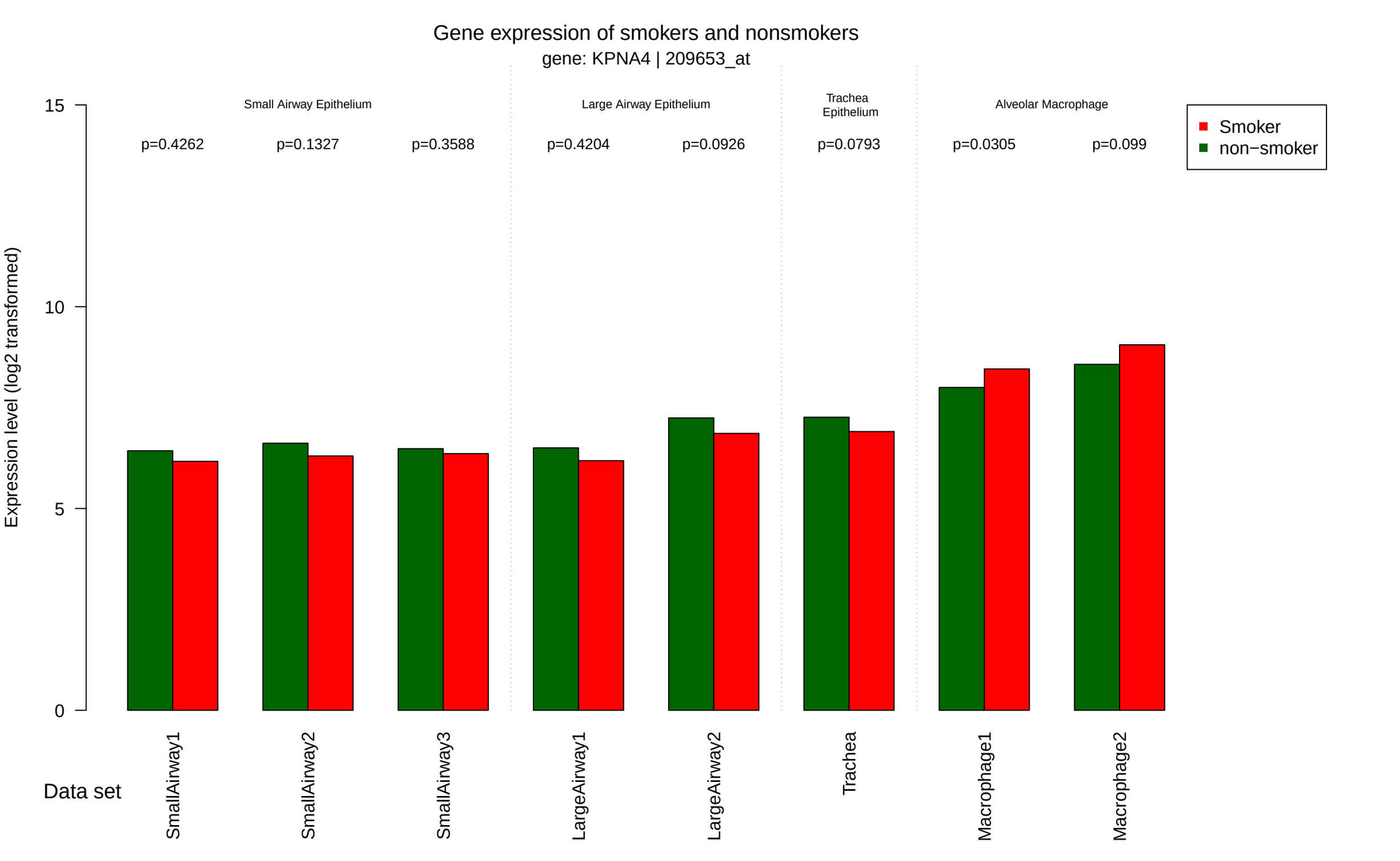
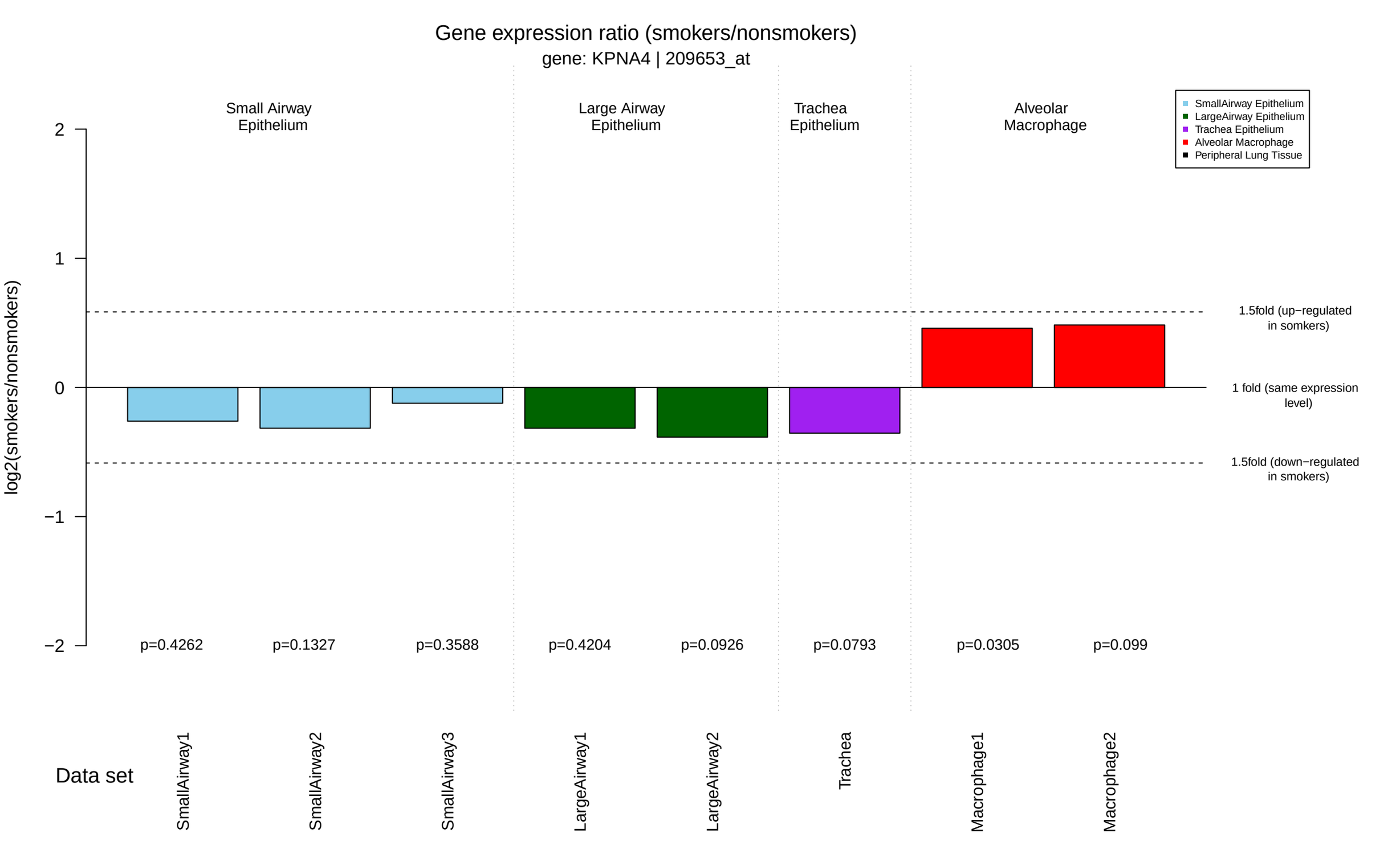
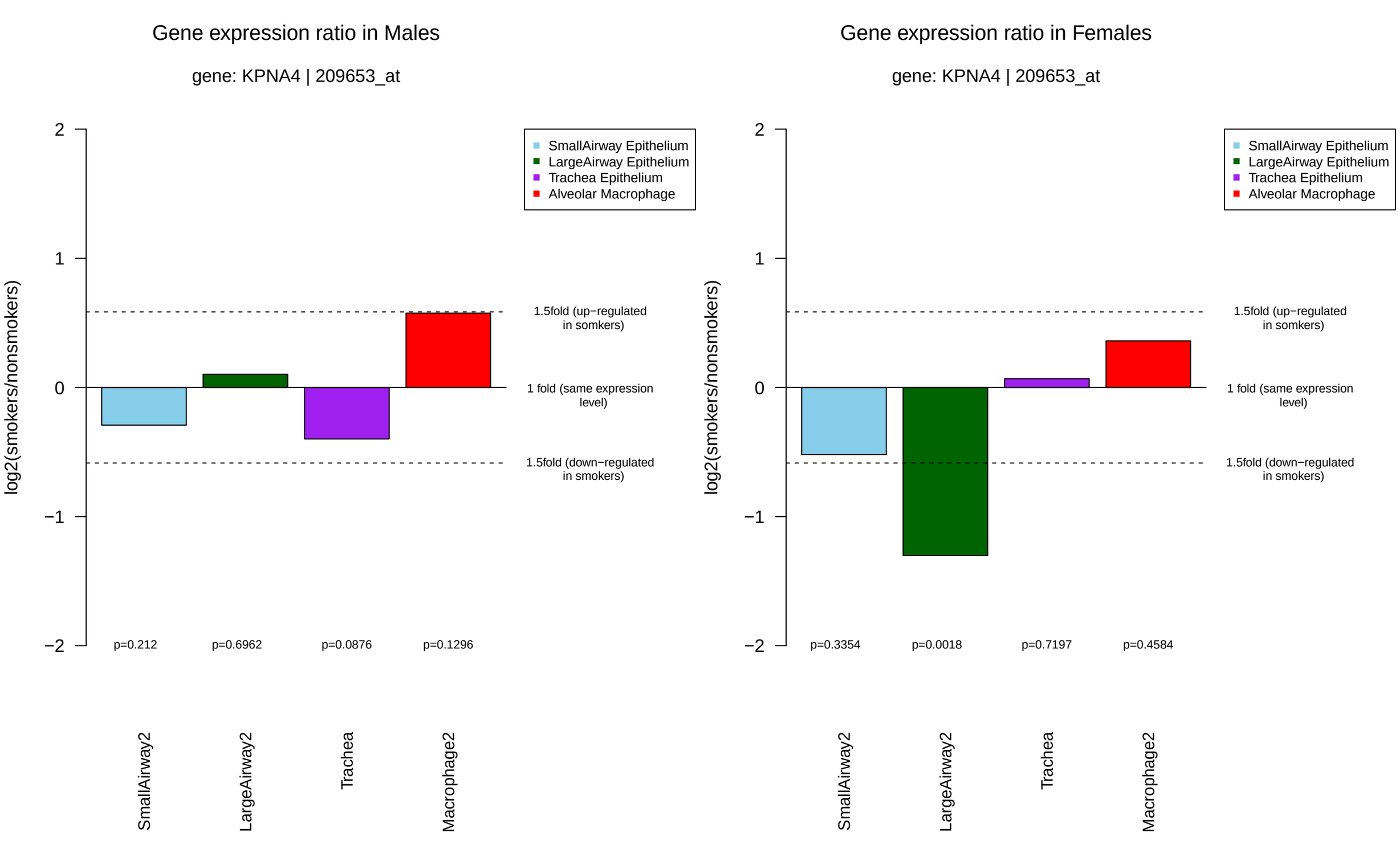
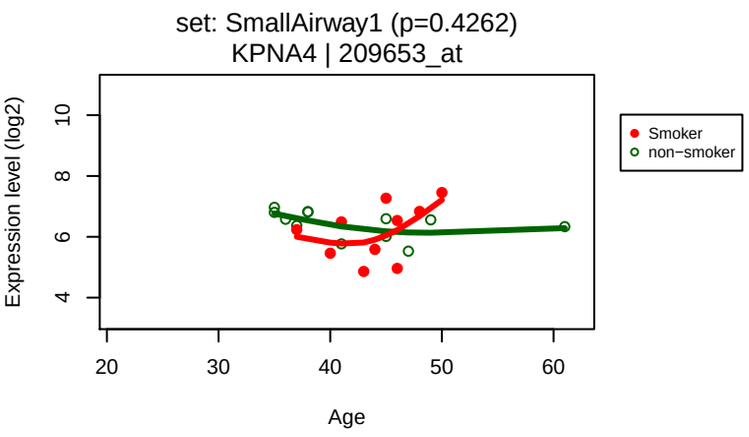
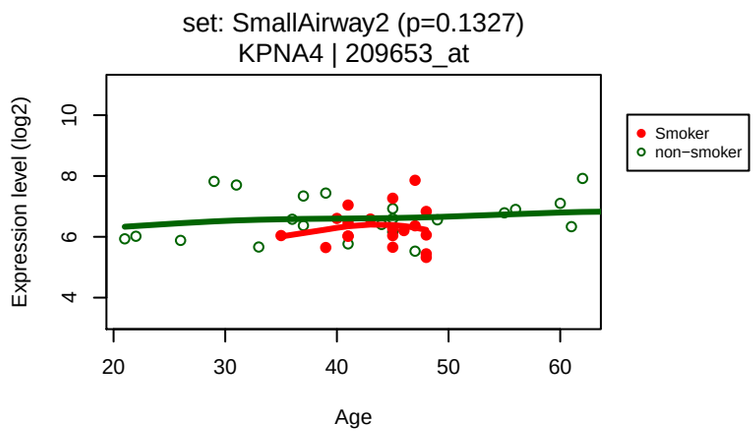
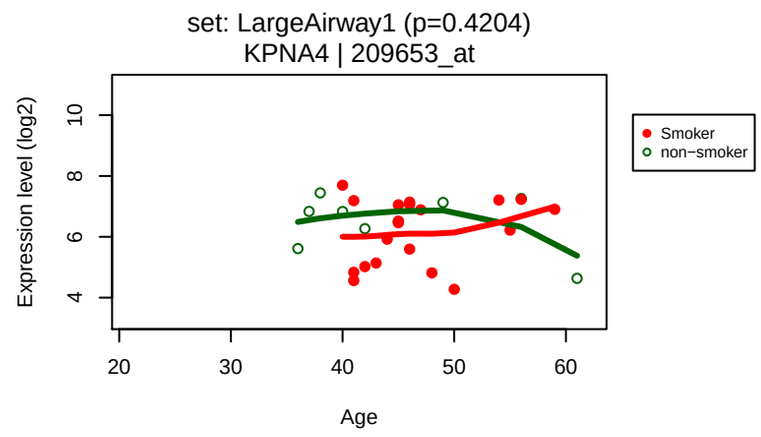

Gene expression of smokers and nonsmokers
gene: KPNA4 | 209653_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4262
p=0.1327
p=0.3588
p=0.4204
p=0.0926
p=0.0793
p=0.0305
p=0.099
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

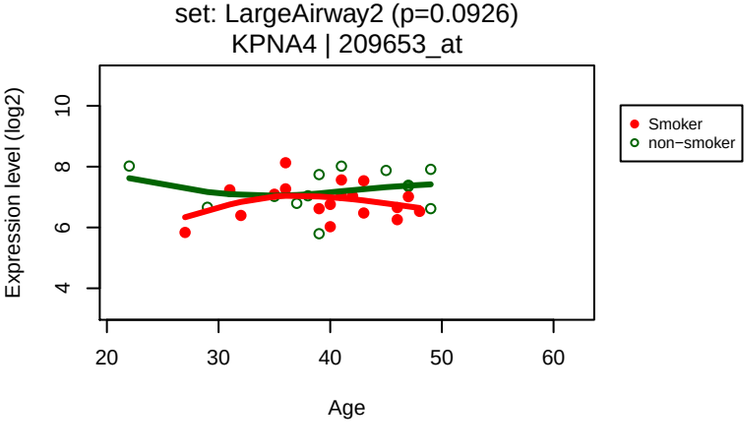

Gene expression ratio (smokers/nonsmokers)
gene: KPNA4 | 209653_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4262
p=0.1327
p=0.3588
p=0.4204
p=0.0926
p=0.0793
p=0.0305
p=0.099
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

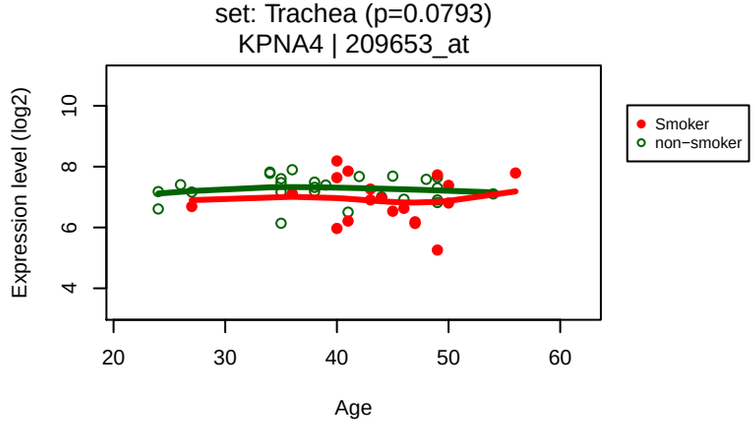

Gene expression ratio in Males
gene: KPNA4 | 209653_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.212
p=0.6962
p=0.0876
p=0.1296
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KPNA4 | 209653_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3354
p=0.0018
p=0.7197
p=0.4584
SmallAirway2
LargeAirway2
Trachea
Macrophage2

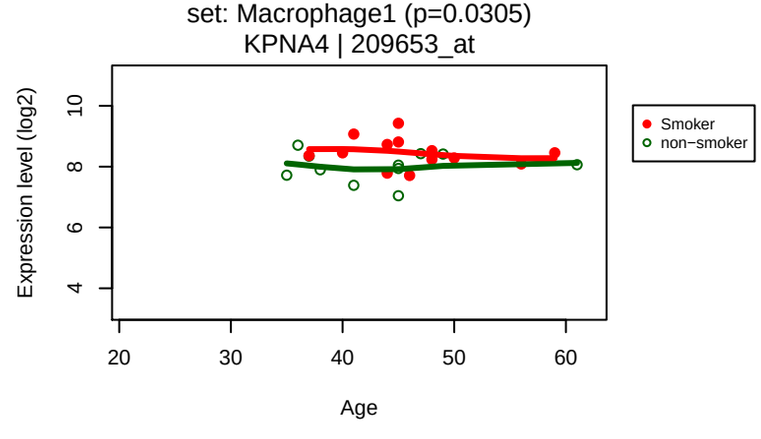

set: SmallAirway1 (p=0.4262)
KPNA4 | 209653_at
Expression level (log2)
Age
Smoker
non−smoker

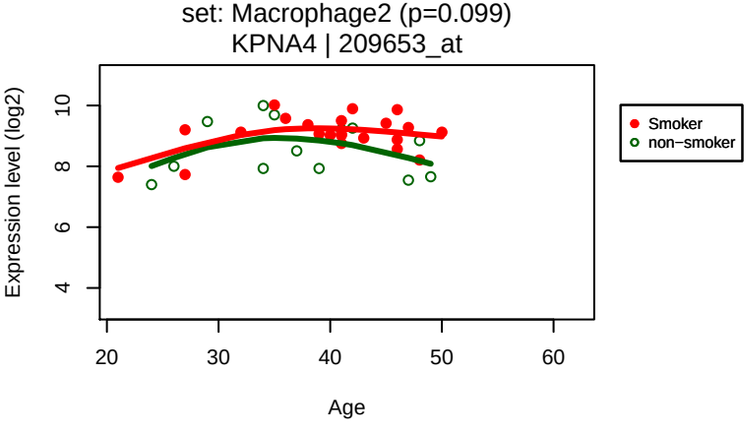

set: SmallAirway2 (p=0.1327)
KPNA4 | 209653_at
Expression level (log2)
Age
Smoker
non−smoker

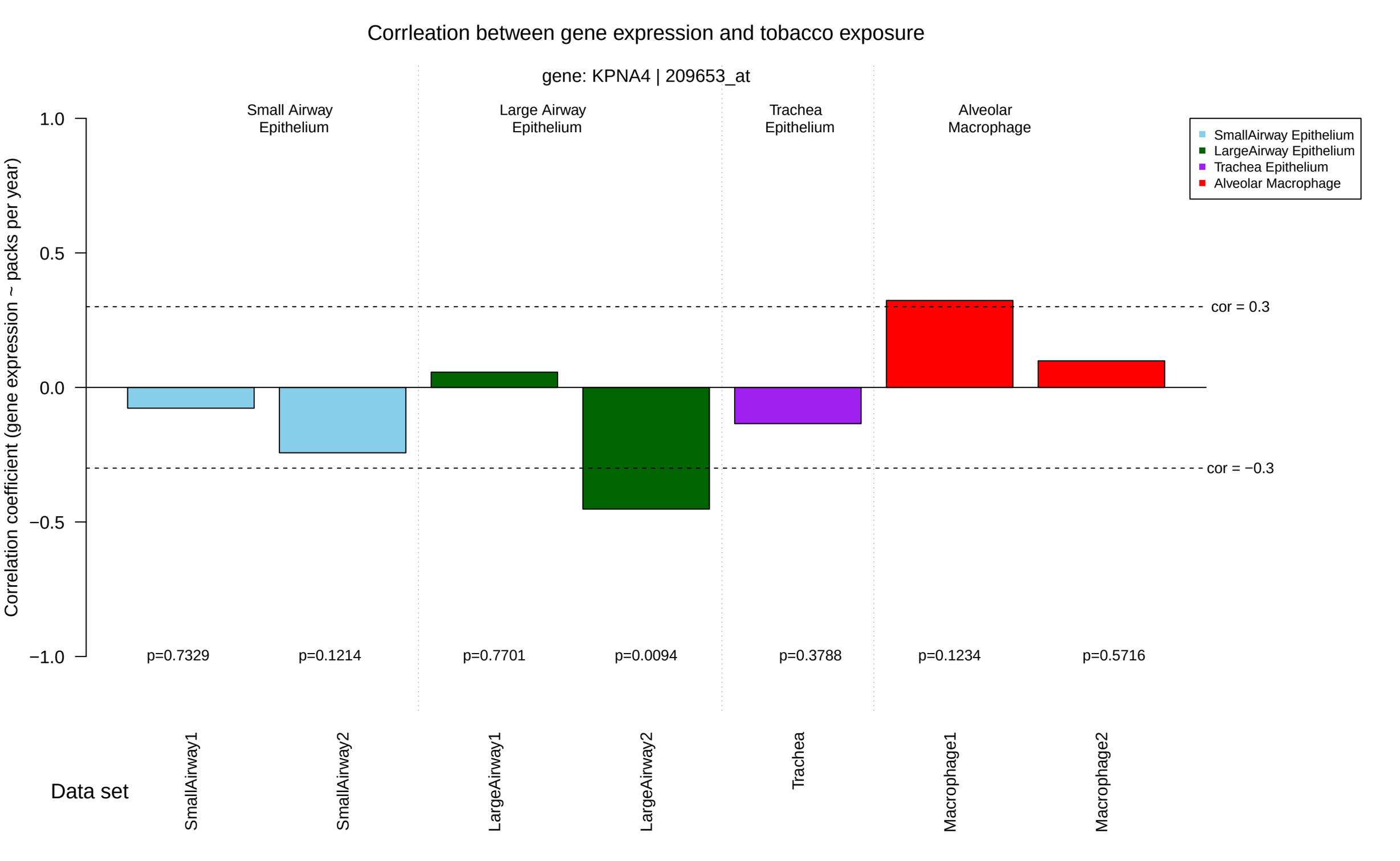

set: LargeAirway1 (p=0.4204)
KPNA4 | 209653_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0926)
KPNA4 | 209653_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0793)
KPNA4 | 209653_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0305)
KPNA4 | 209653_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.099)
KPNA4 | 209653_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KPNA4 | 209653_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7329
p=0.1214
p=0.7701
p=0.0094
p=0.3788
p=0.1234
p=0.5716
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2